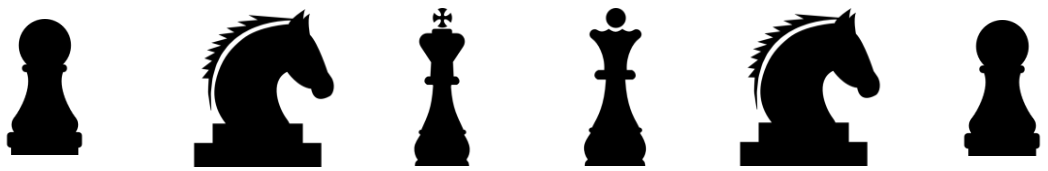
SmoL Chess

Raihan Arefin
2107065

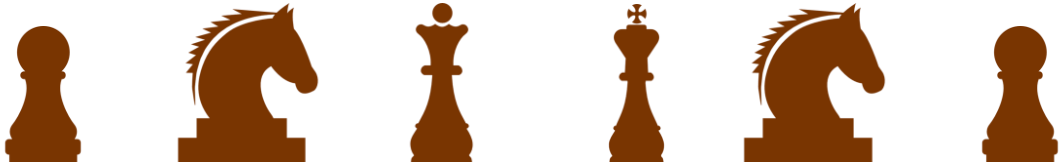
Hassan Mohammed Naquibul Haque
2107077

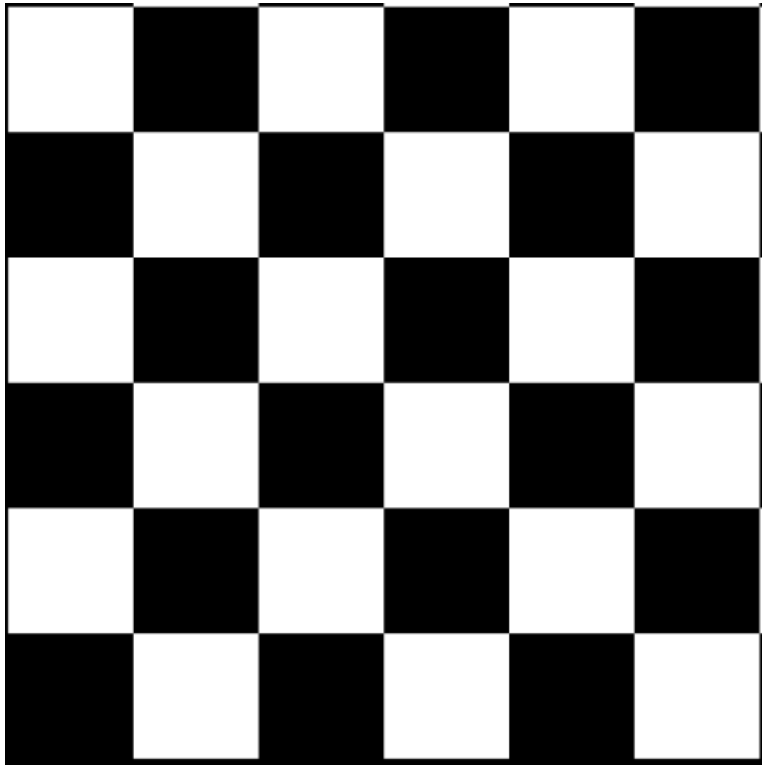
Chess Board Layout



Player 1

Player 2





Chess Board Layout

Piece Moves

The king is the most important piece. Each player has one king, placed near the center of the back row. It moves one square in any direction but cannot move into attacked squares. Protecting the king is critical, and if it cannot escape check, the game ends in checkmate.

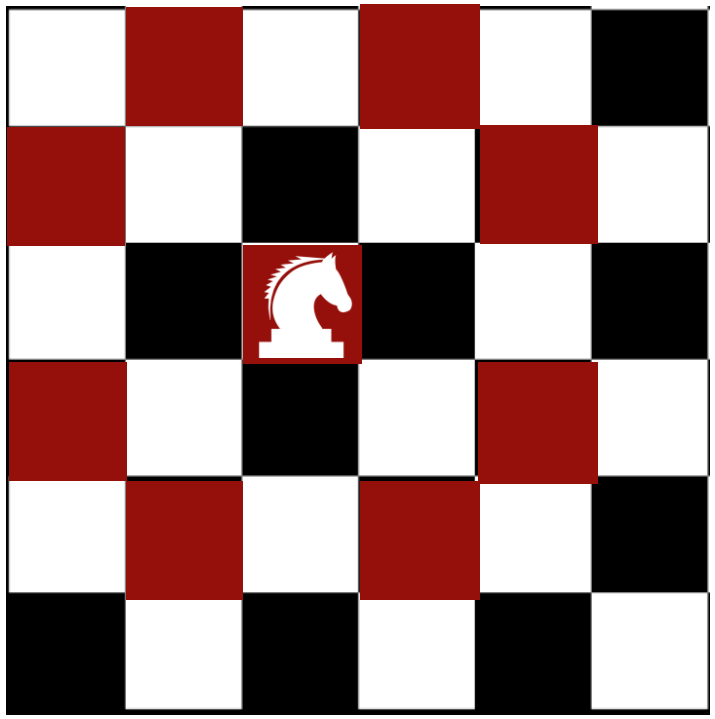
The queen is the strongest piece but range-limited. Each player has one queen next to the king. It moves up to two squares horizontally, vertically, or diagonally. The queen attacks and controls multiple directions but cannot dominate the small board like a normal queen.

Pawns are the smallest pieces on the 6×6 board. Each player has two pawns on the left and right sides. Pawns move one square forward and capture diagonally. They do not have a two-step initial move. When a pawn reaches the last row, it is promoted to a queen. Pawns help control space and support stronger pieces.

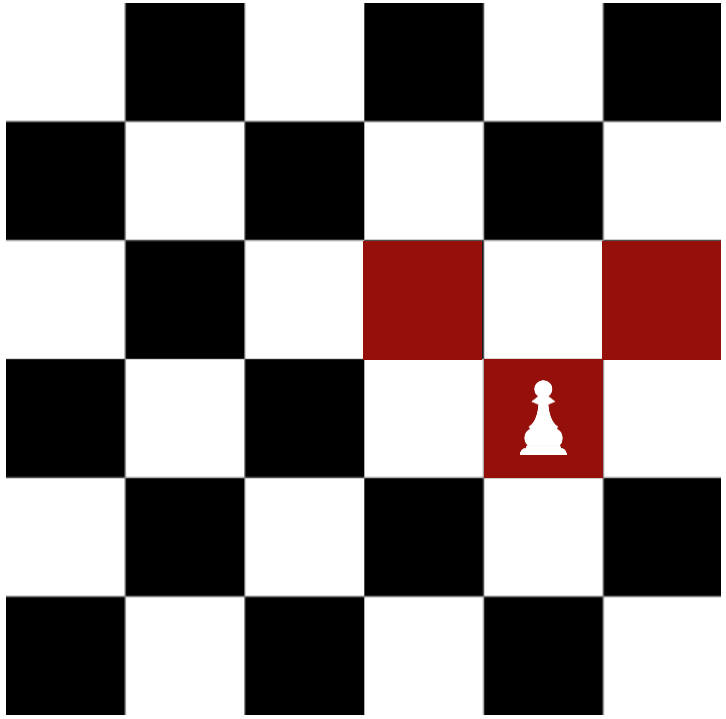
Knights move in an L-shape: two squares in one direction and one perpendicular. Each player has two knights, positioned next to the pawns. Knights can jump over other pieces, making them strong for attacks and forks. On a small board, they can quickly reach key positions.

Chess Board Layout

Knight Moves

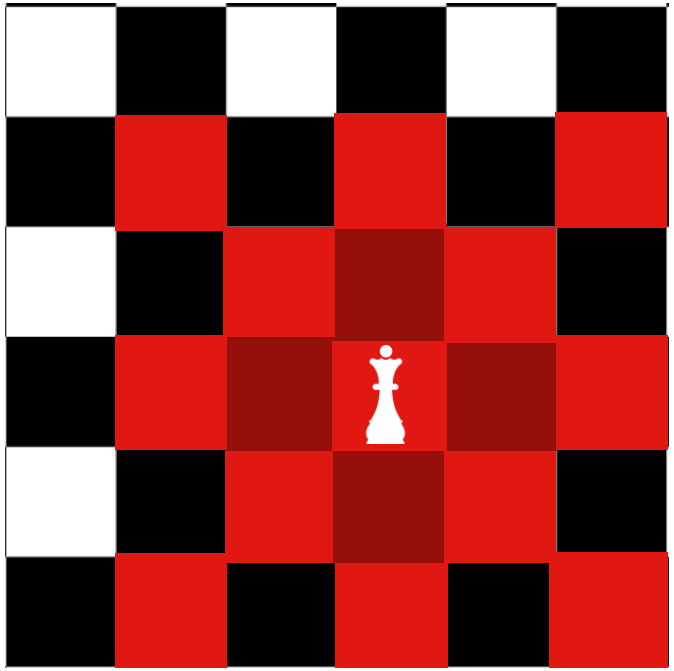


Pawn Moves

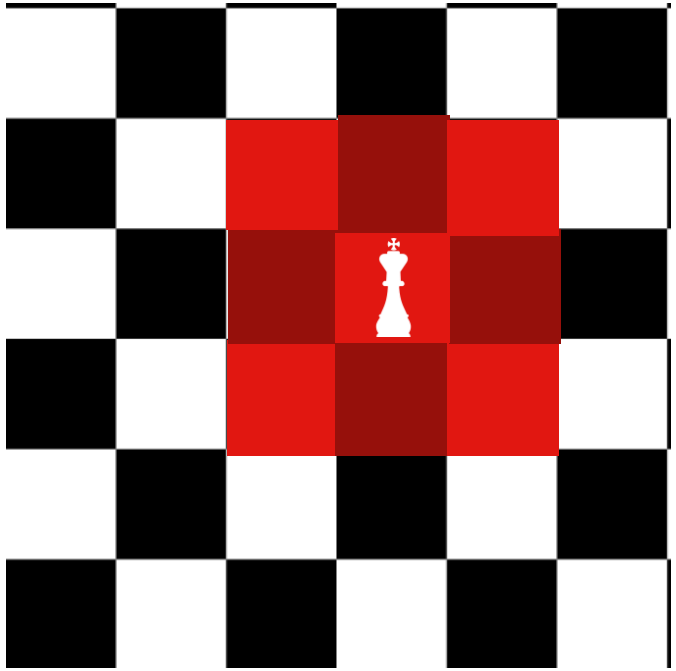


Chess Board Layout

Queen Moves



King Moves



Player 1

Minimax and Alpha-Beta Pruning

Algorithm Strategy:

- Depth-First Search with Alpha-Beta Pruning.
- Iterative Deepening within a time budget.

Evaluation Function (Heuristic):

- Material Score: King (Invaluable) > Queen > Knight > Pawn.
- Mobility: Count of available legal moves.
- Positional Factors: King safety, Center control.

Player 2

Fuzzy Logic System

Input Variables (Fuzzified):

- Material Advantage (Low/Med/High)
- King Danger (Safe/Exposed)
- Center Control & Mobility

Decision Rules (Inference):

- IF King Danger is High THEN Prioritize Defensive moves.
- IF Material Advantage is High AND Danger is Low THEN Simplify (Trade pieces).
- IF Center Control is Weak THEN Favor moves increasing central presence.

Performance Comparison

Minimax Player	Fuzzy Logic Player
Strengths: • Tactical Precision • Deep lookahead	**Strengths:** • Fast computation • Interpretable strategy
Weaknesses: • Slower per move • Calculation heavy	**Weaknesses:** • Short-sighted • Relies on rule tuning

Thank You!